

List of questions for tutorial on Applications of Electromagnetics:

Write answers with neat diagrams and equations (whenever applicable)

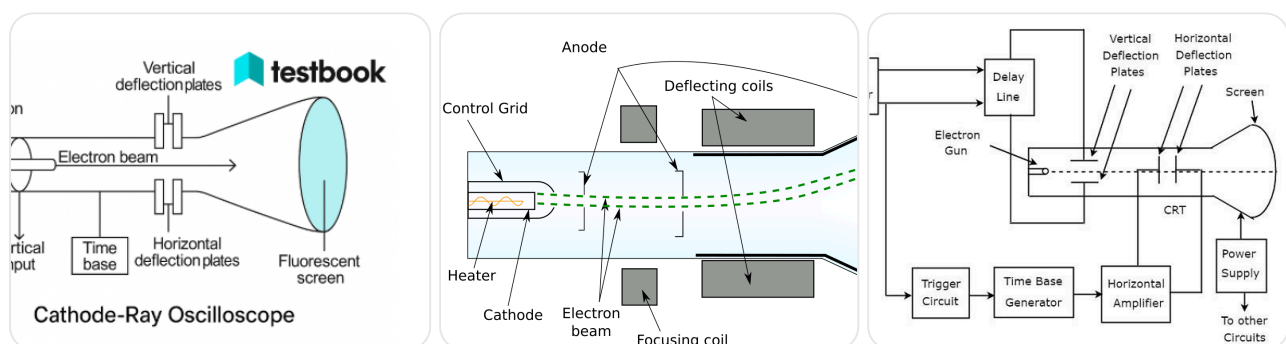
1. Describe the operation of a Cathode Ray Oscilloscope (CRO) using electrostatic deflection
2. Discuss the biological effects of electromagnetic fields on human health
3. Explain different types of antennas and their radiation mechanisms.
4. Describe the working of an electrostatic generator and

Show more



Here are clear, exam-ready answers with diagrams + equations for your *Applications of Electromagnetics* tutorial:

1. Cathode Ray Oscilloscope (CRO) using Electrostatic Deflection



Working Principle

A CRO (Cathode Ray Oscilloscope) uses an electron beam that is deflected by **electric fields** to display signals.

Main Parts

- Electron gun (cathode, grid, anode)
- Vertical deflection plates (Y-plates)
- Horizontal deflection plates (X-plates)
- Fluorescent screen

Operation

1. Electron gun emits electrons → forms a beam.
2. Beam accelerates and strikes the screen.
3. Input signal → applied to **Y-plates** → vertical deflection.
4. Time base signal → applied to **X-plates** → horizontal sweep.
5. Trace on screen represents waveform.

Equation of Deflection

$$\text{Deflection } D \propto \frac{V_d}{V_a}$$

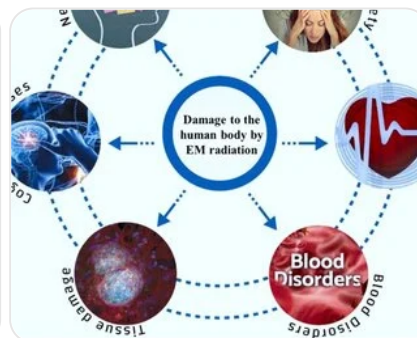
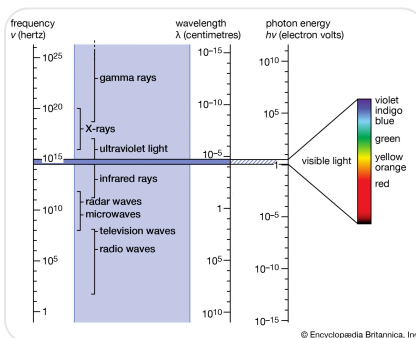
More precisely:

$$D = \frac{e V_d L l}{m v^2 d}$$

Where:

- V_d = deflection voltage
- V_a = accelerating voltage
- L = distance to screen
- l = plate length
- d = plate separation

2. Biological Effects of Electromagnetic Fields



NON-IONISING				IONISING
Extremely low frequency		Radio	Infrared	Ultraviolet
		Microwave	Visible Light	X-ray
				Gamma Rays
Non-thermal		Thermal	Optical	Broken Bonds
Induces Low Currents		Induces High Currents	Excites Electrons	Damages DNA
??? Static Field		Heating	Photo Chemical Effects	
Power Line		AM Radio	FM Radio	Wi-Fi
				Tanning Booth
				Medical X-ray

Types of EM Radiation

- **Non-ionizing:** Radio, Microwave, Infrared
- **Ionizing:** X-rays, Gamma rays

Effects on Human Health

(A) Thermal Effects

- Tissue heating (microwaves)
- Burns at high exposure

(B) Non-Thermal Effects

- Possible DNA damage
- Sleep disturbances
- Headaches

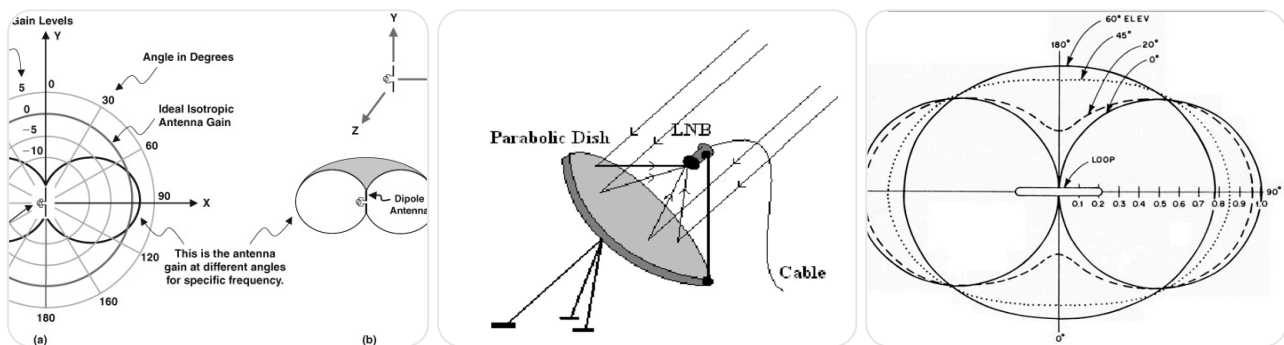
(C) Ionizing Radiation Effects

- Cell mutation
- Cancer risk
- Radiation sickness

Safety Measures

- Limit exposure duration
- Maintain safe distance
- Follow ICNIRP guidelines

3. Types of Antennas and Radiation Mechanism



Types of Antennas

1. Dipole Antenna
2. Monopole Antenna
3. Loop Antenna
4. Parabolic Reflector
5. Horn Antenna

Radiation Mechanism

- Accelerating charges produce EM waves.
- Time-varying current → creates changing electric & magnetic fields.

Key Equation

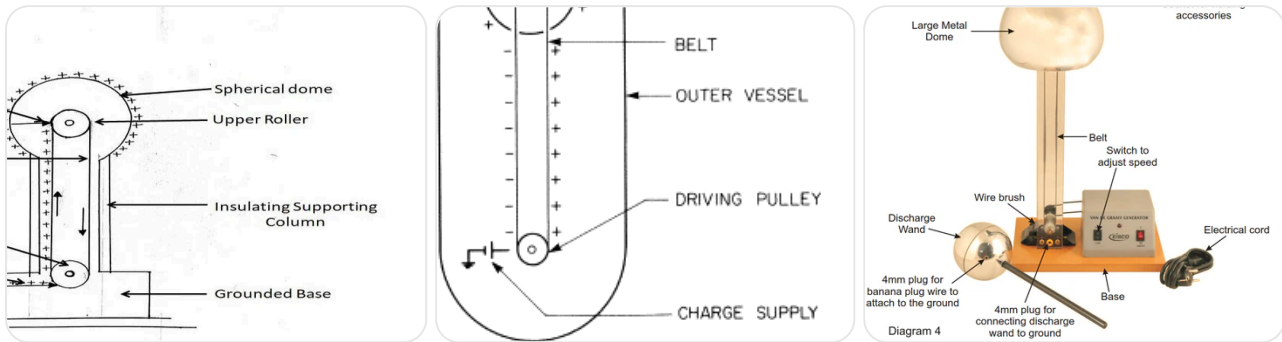
From Maxwell's equations:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Radiation Pattern

- Dipole: Figure-8 pattern
 - Isotropic: Spherical (ideal)
-

4. Electrostatic Generator



Example: Van de Graaff Generator

Working Principle

- Based on **electrostatic charge accumulation**
- Moving belt transfers charge to metal dome

Operation

1. Motor drives insulating belt
2. Charge sprayed onto belt
3. Charge transferred to dome
4. High voltage builds up

Equation

$$V = \frac{Q}{C}$$

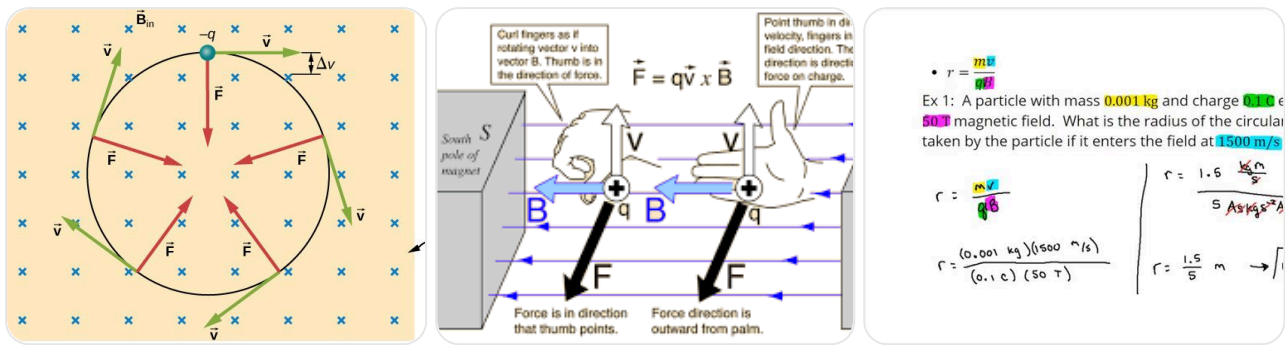
Where:

- Q = charge
- C = capacitance

Applications

- Particle accelerators
 - Nuclear research
 - Electrostatic experiments
-

5. Magnetic Deflection of Charged Particles



Lorentz Force

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$$

Magnitude

$$F = qvB \sin \theta$$

For perpendicular motion:

$$F = qvB$$

Circular Motion

Magnetic force provides centripetal force:

$$qvB = \frac{mv^2}{r}$$

Radius of Path

$$r = \frac{mv}{qB}$$

Time Period

$$T = \frac{2\pi m}{qB}$$

Conclusion

- Particle moves in **circular path**
- Radius depends on m, v, q, B

If you want, I can convert this into a clean PDF or handwritten-style notes for submission





Thanks

You're welcome!

Good luck with your tutorial and submissions. If you want help with:

- handwritten-style answers,
- shorter 5-mark versions,
- viva questions,
- derivations,
- or PDF notes,

just send the topic.

